Citadel-Rx: Continuum-Memory Test-Time Learning for Neural Receivers With Physics-Gated Consolidation Against Pilot Poisoning

Abraheem Rashid

***Abstract*—Neural receivers that learn at test time adapt to the channel from the pilots of every coherence block, so the pilot stream is a training set that an adversary can write to. A perturbation shaped through the receiver's own adaptation step and injected on the pilots destroys online and meta-learned DeepSIC receivers, and meta-learning retains the poison. This paper proposes Citadel-Rx, a DeepSIC receiver whose adaptable parameters are kept at three nested time-scales: a fast level fitted on every block and two slow levels updated every 8 and 32 blocks through a gated average of the faster states. Three label-free physics-consistency gates, each a ratio against a model-based reference on the same block, decide what may be consolidated: a residual test against the linear channel model, a CRC pass-rate test whose threshold follows from the post-adaptation symbol error probability, and an innovation test against the channel prior. A rejected block is reverted and its data are detected with the slow memory. For the algorithm as implemented, the displacement that a budget-ϵ attack causes in the slowest level over W blocks is proved to be bounded by the fast level's displacement times $1-(1-\eta^{(K)})^{\lceil W/C_K\rceil}$, and the bound is verified with measured constants. In 1 926 recorded runs over four channel families, 4×4 and 8×8 systems, three Doppler rates and five attackers, Citadel-Rx keeps its clean symbol error rate at every budget from -10 to $+10$ dB signal-to-interference ratio (attack-window SER 0.026–0.034 against 0.28–0.97 for online DeepSIC and 0.24–0.86 for Meta-DeepSIC; all paired differences significant after Holm correction), rejects 86–99% of poisoned blocks with at most 1.7% false rejections, and, without any attack, halves the symbol error rate of online DeepSIC at 10 dB because the slow memory acts as a prior for the per-block fit. Ablations attribute the clean gain to the proximal anchor and the robustness to the slow memory and the fallback. The limits of the defence are established as well: at symbol error rates near 10^{-4} the poisoned blocks that pass the gate set a floor near 10^{-3}, and on a nearly diagonal channel the clean gain over LMMSE vanishes.**

***Index Terms*—Deep receivers, test-time learning, adversarial pilot poisoning, meta-learning, continual learning, physics-informed gating, MIMO detection, 6G security.**

I. Introduction

DEEP receivers replace or augment the model-based blocks of a MIMO detector with neural networks trained on data [1]–[4]. Since a network trained in simulation meets a different channel in the field, the strongest of them adapt at test time: online DeepSIC [5] re-trains its per-user networks on the pilots of every coherence block, and Meta-DeepSIC [6] meta-learns an initialisation from the pilots of past blocks so that the per-block fit converges faster. Their adaptation data are the only labelled symbols the receiver ever sees: a few hundred pilot bits per user per block.

The security problem addressed here is the following. An adversary within range of the base station adds a perturbation to the received pilot part of a block, with an energy budget expressed as a signal-to-interference ratio (SIR) over the pilots. Wu *et al.* [7] showed that a perturbation shaped through an unrolled copy of the receiver's adaptation step raises the post-adaptation symbol error rate (SER) of online DeepSIC by an order of magnitude, that a perturbation crafted on a surrogate receiver still transfers, and that Meta-DeepSIC, which keeps the poisoned pilots in its support set, does not recover after the attack stops. They call for receivers that are secure by construction; none has been reported. Four properties of the problem shape the design and the evaluation presented here. First, the receiver cannot verify its adaptation data, because the pilots are the labels; what it can verify is physical consistency: whether its decisions fit a linear channel model, whether the CRC of the decoded data passes, and whether the estimated channel moved as a channel does. Second, the harm of poisoning lies not in the attacked block alone but in what the receiver keeps, so a defence must protect the memory and not only the decision. Third, an attacker who knows the defence will try to pass its tests, so the tests must be tied to the attacker's own objective. Fourth, the value of a defence can only be established against the attacks of the literature, under the same channel realisations as the baselines, over many seeds, with the clean cost of the defence reported alongside its benefit.

Prior work addresses these properties separately (Table I). Test-time adaptation in vision has its own poisoning problem [8], and its defences replace batch statistics or gradients by robust estimates [9]; they assume i.i.d. batches and, as shown below, do not transfer to a receiver that sees one correlated block at a time. Byzantine-robust aggregation [10] needs many gradient vectors per step, which a receiver with one pilot block does not have. Certified defences against data poisoning [11] bound the loss of a convex learner under a fraction of corrupted points; the present setting is sequential, the learner is a network, and the corruption is a bounded perturbation of every point in a block. Corruption-robust online learning [12] supplies the additive accounting used in Section IV. Nested learning [13] and Titans [14] keep memory at several update

A. Rashid is with the Department of Computer Science, Brunel University London, London, U.K. (e-mail: 2546393@brunel.ac.uk).

Code, configuration, every run record and the scripts that regenerate every table and figure are available at https://github.com/abraheemrashid99-sys/Citradel-Rx.

TABLE I
POSITIONING BY METHOD CLASS. THE ATTACK SURFACE IS WHAT THE ADVERSARY CAN WRITE TO; –: NOT ADDRESSED.

Class	Adapts	Memory	Attack surface / defence
LMMSE	no	–	pilot estimate / –
Online DeepSIC [5]	every block	last block	pilots / –
Meta-DeepSIC [6]	every block	pilot buffer	pilots, persistent / –
Drift-detection receiver	on drift	last block	pilots / adapt less
Robust TTA, vision [8], [9]	every batch	batch statistics	batch / robust statistics
Byzantine aggregation [10]	–	–	gradients / needs many vectors
Nested, continuum memory [13], [14]	every token	multi-rate	input / –
Citadel-Rx (this paper)	every block	three rates, gated	pilots / physics gates, certified retention

TABLE II
SIMULATION SETUP, READ FROM THE CONFIGURATION FILES AND THE CODE.

Item	Value
System	$U \times N = 4 \times 4$ (8×8 in Section VI-I), QPSK, SNR 10 dB (swept 4–16 dB)
Block	$N_p = 100$ pilot symbols (200 bits) and $N_d = 25\,000$ data symbols per user
CRC	CRC-8 with the 3GPP polynomial [22]; $L = 32$ symbols per sub-block; $n = 781$ sub-blocks per block
Channel	SED $[\boldsymbol{H}]_{ij} = e^{-\lvert i-j\rvert}$ with per-user cosine fading (periods 51, 39, 33, 21 blocks; depth 0.8) [5]; Jakes block correlation $\rho = J_0(2\pi f_D T_b)$ with $f_D T_b \in \{0.01, 0.05, 0.1\}$ [20]; $\tanh$ PA nonlinearity; COST2100 traces released with [6] [21]
Run	100 blocks (400 for A3); attack on blocks 30–69; seeds 0–2 (0–4 on the headline cells)
DeepSIC	$Q = 3$ iterations; per-user MLP with $64{\cdot}$order hidden units and ReLU [5]
Adaptation	Adam, learning rate $5 \cdot 10^{-3}$, 300 epochs per block on the pilots; joint pre-training at 2–16 dB
Attacker	PGD with 20 steps of size $0.25\,\epsilon$ through 2 unrolled adaptation steps [7], [16]

frequencies; they do not decide what enters the slow memory, and a single test-time memory is, as the ablation shows, as vulnerable as online DeepSIC. Table I summarises the gap: no published receiver keeps memory at several rates, gates what enters the slow rates on physics the attacker cannot forge without giving up its objective, and carries a retention guarantee checked against measurements.

The contributions of this paper are as follows.

1) **A three-timescale receiver** (Section III): a fast level with a delta-rule learner and a surprise-generated step size, slow levels at periods aligned with the coherence structure of the channel, gated consolidation, a proximal anchor that pulls the fast level toward the slow projection, and a gated fallback that detects a rejected block with the uncorrupted memory.
2) **Relative physics gates** (Section III-D): three label-free tests whose statistics are ratios against a model-based reference computed on the same block, so that a channel that merely degrades does not look like a poisoned adaptation. The CRC gate's threshold is derived, not tuned, from a target excess SER (Proposition 1), and a relative slack keeps it meaningful at 10^{-3} operating points.
3) **A retention certificate for the implemented algorithm** (Section IV): the exposure of the slowest level to a bounded attack is divided by its period and multiplied by the miss rate of the gate (Theorem 1), with a regret decomposition in which the adversary's term is additive (Theorem 2). Every constant is measured and the bound is drawn against paired adversarial/clean memory trajectories.
4) **A reproducible, paired evaluation** (Sections V–VI): seven receivers, five attackers, four channel families, two system sizes, three Doppler rates and three to five seeds; bootstrap intervals and Holm-corrected Wilcoxon tests on per-block differences paired on the channel realisation; every number regenerated from 1 926 recorded runs by one script each.

The evaluation also establishes where the defence does *not* help: at clean SER near 10^{-4} the attacked window sits near 10^{-3}, because the few poisoned blocks that pass the gate dominate; on a nearly diagonal channel the clean gain over LMMSE vanishes; and the first calibration of the relative CRC slack raised false rejections to 15–20% before it was corrected. Section VI-H attributes the advantages of the receiver to its individual components.

Section II presents the system and threat model, Section III the receiver, Section IV the guarantees, Section V the experimental setup, Sections VI and VII the results and their scope, and Section VIII concludes.

II. SYSTEM MODEL AND THREAT MODEL

This section fixes the uplink, the receiver under attack, the attacker and the metrics by which every method is judged.

A. Uplink and Adaptive Receiver

U single-antenna users transmit QPSK symbols to a base station with N antennas. Time is divided into coherence blocks $b = 1, 2, \ldots$; within block b the channel $\boldsymbol{H}_b \in \mathbb{C}^{N \times U}$ is constant and the received vector at symbol slot n is

$$\boldsymbol{y}_n = f(\boldsymbol{H}_b \boldsymbol{s}_n) + \boldsymbol{w}_n, \qquad \boldsymbol{w}_n \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}), \tag{1}$$

with f the identity (linear channel) or the memoryless power-amplifier nonlinearity $f(x) = \tanh(0.5x)$ applied per antenna, as in [6]. Each block carries N_p pilot symbols per user, known to the receiver, followed by N_d data symbols framed in CRC-protected sub-blocks of L symbols (Table II). Four channel families are used (Section V).

The receiver is DeepSIC [5]: Q soft-interference-cancellation iterations, each a per-user multilayer perceptron that maps the received vector and the other users' current soft decisions to a posterior over the user's symbol. Its parameters θ are adapted on the pilots of every block,

$$\theta_b = \Phi(\theta_{b-1}; \boldsymbol{Y}_b), \tag{2}$$

by T steps of a first-order optimiser on the pilot cross-entropy, and then detect the data of the block. Online DeepSIC starts every block from θ_{b-1}; Meta-DeepSIC starts from an initialisation meta-learned by MAML [15] on a buffer of past pilot blocks, refreshed every five blocks [6].

B. Threat Model

The adversary adds $\boldsymbol{E}_b$ to the received pilot matrix $\boldsymbol{Y}_b \in \mathbb{C}^{N\times N_p}$ of block b, with $\|\boldsymbol{E}_b\|_F \le \epsilon_b$ set by the SIR over the pilot block,

$$\epsilon_b = \|\boldsymbol{Y}_b\|_F \, 10^{-\mathrm{SIR}/20}, \tag{3}$$

so that SIR $0\,$dB spends as much energy as all U users together; the data part is not touched. The attacker knows the pilots, which are standardised, and observes the channel through its own receiver. Following and extending [7], four attackers are considered.

- *A1, availability.* Projected gradient ascent [16] on the post-adaptation data loss, differentiated through an unrolled copy of the victim's adaptation map (2) (white-box), or crafted on a surrogate online DeepSIC and injected (transfer). A receiver without an adaptation map (LMMSE) is attacked through the surrogate.
- *A2, targeted drift.* A1 with an objective that raises one user's loss while keeping the others' low, on a Jakes channel, so that the poison resembles a plausible channel change.
- *A3, boiling frog.* Many blocks of small perturbations (5% of the pilot energy) that push the receiver toward a spoofed channel basin while keeping the SER inside any tolerance a detector might use.
- *Gate-aware.* A1 with a penalty on the defence's gate statistics in the attacker's objective: an attacker who knows the defence and tries to pass it.

A *data-jamming control* spends the same energy on the data part of the block instead of the pilots; it separates persistent poisoning from plain interference.

C. Metrics

The SER is the fraction of symbol slots in which at least one user is wrong, the metric of [5]–[7], reported as a mean over the attack window (blocks 30–69), before it (pre) and after it (post). The recovery time is the number of blocks after the attack until the 3-block moving average of the SER returns within 10% of the pre-attack mean (30: not within the run). The gate is characterised by its rejection rate of poisoned blocks and its false-rejection rate of clean blocks. The degradation Δ_{dB} is the extra SNR needed under attack to reach the clean SER [7]. The paired displacement $D_k(W) = \left\|\theta^{(k)}_{\mathrm{adv}} - \theta^{(k)}_{\mathrm{clean}}\right\|$ of memory level k is measured between an attacked and a clean run on the same channel. Fig. 1 summarises the study from configuration to every reported number.

III. Citadel-Rx

The receiver splits adaptation by time-scale: a fast level fits every block; slow levels are updated at the periods of the coherence structure of the channel, and only with what the gates accept. Fig. 2 shows the architecture.

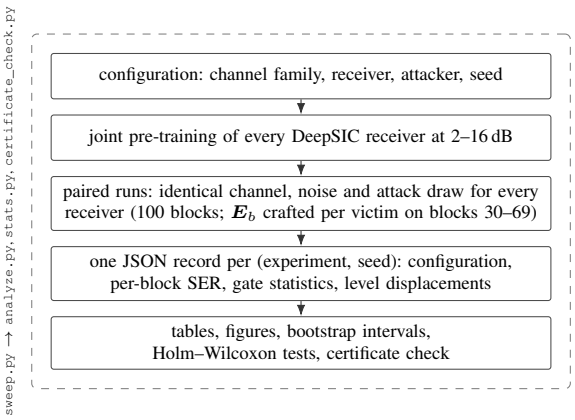


Fig. 1. Study pipeline from configuration to every reported number. Finished records are skipped on re-run, so every stage is resumable.

A. Levels, Periods and the Fast Learner

The adaptable parameters $\theta \in \mathbb{R}^d$ (all DeepSIC weights, $d = 15\,628$ for 4×4) are kept at $K = 3$ levels $\theta^{(1)}, \theta^{(2)}, \theta^{(3)}$ with periods $C = (1, 8, 32)$ blocks, each a full copy. The fast level is adapted on every block by $T = 300$ steps of

$$\theta \leftarrow \theta - \eta_b \delta\, \theta\, \boldsymbol{\phi}\boldsymbol{\phi}^{\top} \quad \text{(delta-rule decay, per weight matrix)}, \tag{4}$$

$$\theta \leftarrow \theta - \eta_b \lambda(\theta - a_b) \quad \text{(proximal pull to the anchor)}, \tag{5}$$

$$\theta \leftarrow \mathrm{Adam}\big(\theta, \nabla_\theta \ell_b(\theta)\big), \tag{6}$$

where ℓ_b is the pilot cross-entropy of the DeepSIC stage being adapted (stages are adapted sequentially as in [5]), $\boldsymbol{\phi}$ is the normalised mean input of the stage (the direction along which the delta rule forgets), $\delta = 0.1$, $\lambda = 0.5$, and $a_b = \sum_{k\ge 2} w_k \theta^{(k)}_{b-1}$ with $w = (\frac{1}{2}, \frac{1}{2})$ is the *anchor*, the slow projection. The step size is generated by the receiver's own surprise: with $S_b = \|\nabla_\theta \ell_b(\theta_{b-1})\|^2$ and $\bar{S}_b$ its exponential moving average over past blocks, seeded by the median of the first five,

$$\eta_b = \eta_{\max}\frac{2r_b}{1+r_b}, \qquad r_b = \mathrm{clip}\big(S_b/\bar{S}_b, \tfrac{1}{8}, 8\big), \tag{7}$$

so that an unsurprising block receives a small step and a surprising one at most $2\eta_{\max}$. The rate is bounded, which the analysis requires.

B. Gated Consolidation

Every C_{k+1} blocks, level $k{+}1$ receives the gated mean of the level-k states buffered since its last update, each weighted by the gate verdict $\gamma_j \in [0,1]$ of its block (level 1) or by the mean verdict of its period (level 2):

$$\theta^{(k+1)} \leftarrow \theta^{(k+1)} + s\Big(\frac{\sum_j \gamma_j \theta^{(k)}_j}{\max(1, \sum_j \gamma_j)} - \theta^{(k+1)}\Big), \qquad s = \eta^{(k+1)} \min\big(1, \textstyle\sum_j \gamma_j\big), \tag{8}$$

with $\eta^{(2)} = 1$ (level 2 is a running mean of the fast level) and $\eta^{(3)} = 0.5$. Eq. (8) is a convex combination (Lemma 1): a level never leaves the convex hull of what it has been shown, and

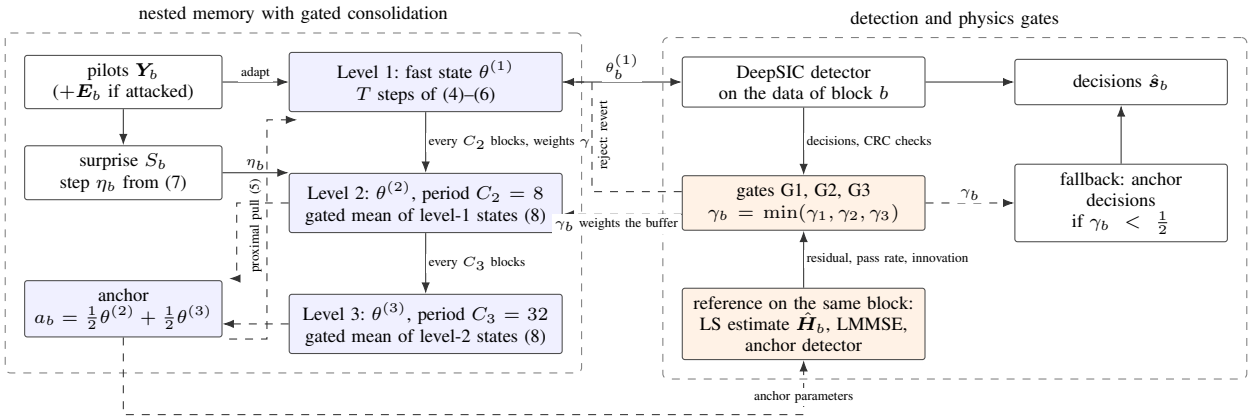


Fig. 2. Citadel-Rx. Solid arrows carry parameters and decisions; dashed arrows carry the gate verdict and the slow projection fed back. Level $k{+}1$ receives the gated mean of level-k states every C_{k+1} blocks (8); the fast level is pulled toward the anchor on every step; a rejected block is reverted and its data are detected with the anchor.

TABLE III
HYPERPARAMETERS, READ FROM THE CONFIGURATION FILE AND THE CODE.

Levels	$K = 3$; periods $C = (1, 8, 32)$; $\eta^{(2)} = 1$, $\eta^{(3)} = 0.5$; anchor weights $(\frac{1}{2}, \frac{1}{2})$
Fast learner	Adam with $\eta_{\max} = 5 \cdot 10^{-3}$, $T = 300$; delta decay $\delta = 0.1$; proximal weight $\lambda = 0.5$
Surprise	EMA weight 0.9; warm-up 5 blocks; clip to $[1/8, 8]$
G1	residual ratio threshold $\tau_1 = 1.25$ on the clean-EMA-normalised statistic
G2	slack $\delta = 0.01$ (absolute) or $\max(2 \cdot 10^{-3}, p_{\mathrm{ref}})$ (relative); $L = 32$
G3	innovation ratio threshold $\tau_3 = 4$, floored by the $\chi^2_{0.999}$ quantile per degree of freedom
Gate	soft step of width 0.1τ; $\gamma = \min$; thresholds tightened by up to 30% under surprise
Rejection	revert the fast update; detect the block with the anchor

a period in which everything was rejected ($\sum \gamma < 1$) moves it proportionally less. The fast level is anchored continuously by (5) rather than re-set at consolidation instants as in [13]; Section VI-H shows that this choice is the source of the clean gain.

C. Rejection, Reversion and Fallback

If $\gamma_b < \frac{1}{2}$ the block is rejected: the fast update is reverted, $\theta_b^{(1)} = \theta_{b-1}^{(1)}$; the reverted state enters the buffer with weight γ_b; and the data of the block are detected with the anchor, the memory the attacker has not reached. Reversion protects the memory; the fallback protects the attacked block itself (Section VI-H: without it the attack-window SER is 0.245, with it 0.028). Table III lists every hyperparameter.

D. Relative Physics Gates

Every statistic is a ratio against a reference computed on the same block from the received data and no labels: the LMMSE detector built on the least-squares pilot channel estimate $\hat{\boldsymbol{H}}_b$, and the detector obtained with the anchor parameters; the reference is the better of the two. Ratios are normalised by their exponential moving average over accepted blocks and compared with a soft threshold, tightened by up to 30% when

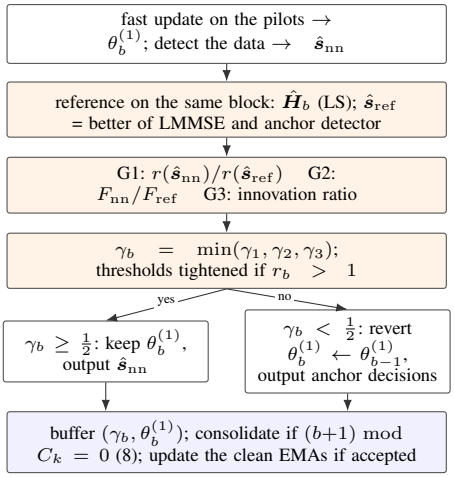


Fig. 3. Decision inside one block. Shaded: gate computations (orange) and memory operations (blue).

the surprise ratio r_b exceeds one. Fig. 3 gives the decision inside one block.

G1, model residual. With $r(\hat{\boldsymbol{s}}) = \frac{1}{N_d}\sum_n \left\| \boldsymbol{y}_n - \hat{\boldsymbol{H}}_b \hat{\boldsymbol{s}}_n \right\|^2$, the statistic is $r(\hat{\boldsymbol{s}}_{\mathrm{nn}})/r(\hat{\boldsymbol{s}}_{\mathrm{ref}})$. A poisoned adaptation produces decisions inconsistent with the linear model. Both terms contain the same noise and model-mismatch energy, so a PA nonlinearity compresses the ratio toward one and cannot raise a false alarm (Proposition 2).

G2, CRC pass rate. The fraction F_{nn} of CRC sub-blocks that pass with the fast level, over the fraction F_{ref} with the reference, against the threshold τ_2 of Proposition 1. This gate is tied to the availability attacker's own objective: an attack that raises the SER lowers the pass rate.

G3, channel prior. The innovation energy of the LS channel estimate against its AR(1) prediction, relative to its running level on accepted blocks and floored by the χ^2 quantile of the degrees of freedom. It targets physically implausible jumps and is the only gate not fooled by an attacker who keeps the SER low.

IV. GUARANTEES

The guarantees are stated for the algorithm as implemented; the full proofs are given in the theory document of the repository. Let $\Phi(\theta; \boldsymbol{Y}, a)$ denote the block adaptation map (the fast state after one block, given the pilots and the anchor), and consider paired runs on the same clean sequence with and without an attack, with displacements $d_b = \left\| \theta_b^{(1),\mathrm{adv}} - \theta_b^{(1),\mathrm{clean}} \right\|$ and $D_k(b)$ for level k.

Lemma 1 (consolidation is a convex combination)**.** *Eq.* (8) *writes the new level-$(k{+}1)$ state as a convex combination of its old state and the buffered level-k states, with total weight $s \le \eta^{(k+1)}$ on the latter. Consequently, if the paired runs have equal verdicts, $D_{k+1} \le \max\{D_{k+1}^{\mathrm{old}}, \max_j D_k(j)\}$, and by induction $D_k(b) \le \max_{j \le b} d_j$ for every $k \ge 2$.*

Theorem 1 (retention certificate)**.** *Let an attack occupy a window of W blocks with equal gate verdicts in the paired runs, let $n(W) \le \lceil W/C_K \rceil$ be the number of level-K consolidations inside it, and let $\Delta_f(W) = \max_{\mathrm{window}} d_b$ be the largest paired fast-level displacement. Then*

$$D_K(W) \le \Delta_f(W)\big(1-(1-\eta^{(K)})^{n(W)}\big) \le \eta^{(K)} \Delta_f(W)\, n(W). \tag{9}$$

If moreover $\|\Phi(\theta; \boldsymbol{Y}+\boldsymbol{E}, a) - \Phi(\theta; \boldsymbol{Y}, a)\| \le G_\Phi \|\boldsymbol{E}\|$ and Φ is L_θ- and L_a-Lipschitz in θ and a with $L = L_\theta + L_a < 1$, then $\Delta_f(W) \le G_\Phi \epsilon \min\{1, (1-L_\theta) n_W\}/(1-L)$, where $n_W = \rho_{\mathrm{blk}}(1-p^{\mathrm{rej}})W$ counts the attacked blocks the gate accepts; if only $L \le 1$ holds (non-expansive map), $\Delta_f(W) \le G_\Phi \epsilon\, n_W$.

Proof sketch. Level K consolidates $n(W)$ times in the window; each time, by Lemma 1, $D_K \leftarrow (1-s)D_K + s\Delta_f$ with $s \le \eta^{(K)}$, which iterated from $D_K = 0$ gives (9). For the second part, the triangle inequality on Φ gives $d_b \le L_\theta d_{b-1} + L_a \max_{k \ge 2} D_k(b-1) + G_\Phi \epsilon \chi_b$ with $\chi_b = 1$ on attacked accepted blocks; with the lemma and the geometric sum this solves to the stated bound, and with $L \le 1$ the same inequality is non-expansive and adds at most $G_\Phi \epsilon$ per accepted attacked block. □

The first part of Theorem 1 needs no regularity of the fast learner: the exposure of the slow level is divided by its period whatever the fast level does. The second part identifies where the gate enters, through n_W, and why rejected blocks inject nothing: they are reverted. The form $\eta^{(K)} \rho\, \epsilon\, G(1-p^{\mathrm{rej}})W/C_K$ holds with $G = G_\Phi/(1-L)$, the Lipschitz constant of the whole block adaptation rather than of a single gradient.

Theorem 2 (corruption-robust dynamic regret)**.** *Let ℓ_b be G_ℓ-Lipschitz in θ and $R_T(u) = \sum_b \ell_b(\theta_b) - \ell_b(u_b)$ the dynamic regret against a comparator sequence. Under the hypotheses of Theorem 1, $R_T^{\mathrm{adv}}(u) \le R_T^{\mathrm{clean}}(u) + G_\ell\, T\, \Delta_f(T)$; without the anchor ($L_a = 0$), $R_T^{\mathrm{adv}}(u) \le R_T^{\mathrm{clean}}(u) + G_\ell G_\Phi \epsilon\, n_T/(1-L_\theta)$; and a detector that uses level k pays at most $G_\ell \Delta_f(T) \min\{T, \eta^{(k)} T^2/(2C_k)\}$.*

The adversary's contribution is additive to the clean regret, linear in the number of accepted corrupted blocks and divided by the level period. No rate is claimed for the clean term of the full-MLP receiver; it is reported empirically. For a last-layer-adaptable (convex) variant, the $O(\sqrt{T(1+P_T)})$ dynamic-regret bound of [17], [18] applies with the bounded step (7).

TABLE IV
CONSTANTS OF THEOREM 1 MEASURED BY FINITE DIFFERENCES ON CLEAN BLOCKS OF THE A3 CONFIGURATION: MAXIMUM OVER 10 BLOCKS AND 3 RANDOM PROBES, MEAN IN PARENTHESES. SCRIPT: CERTIFICATE_CHECK.PY.

λ	G_Φ	L_θ	L_a	$L_\theta + L_a$	ϵ (A3)
0.5 (default)	5.21 (3.13)	0.86 (0.76)	0.35 (0.32)	1.20	1.05
1	1.54 (1.43)	0.54 (0.52)	0.54 (0.53)	1.08	1.11
2	2.22 (1.97)	0.40 (0.35)	0.82 (0.80)	1.22	1.11
4	2.21 (2.06)	0.38 (0.29)	1.01 (0.99)	1.39	1.11

Proposition 1 (CRC gate)**.** *With i.i.d. symbol errors of probability p in a sub-block of L symbols and an r-bit CRC whose syndrome is uniform on error, $P(\mathrm{pass} \mid p) = (1-p)^L + 2^{-r}(1-(1-p)^L)$. Inverting at F_{ref} gives p_{ref}, and the rule "reject if $F_{\mathrm{nn}}/F_{\mathrm{ref}} < \tau_2$" with*

$$\tau_2 = \Big(\frac{1-p_{\mathrm{ref}}-\delta}{1-p_{\mathrm{ref}}}\Big)^L, \qquad \delta = \max\{\delta_{\min}, \rho\, p_{\mathrm{ref}}\}, \tag{10}$$

rejects exactly when the fast level's error probability exceeds the reference's by more than δ, up to the Hoeffding fluctuation $2e^{-2nt^2}$ of the pass fraction over n sub-blocks [19].

With $n = 781$ the gate is deterministic for practical purposes against any attacker that raises the SER by more than δ; an attacker that does not has caused less than δ of harm. The absolute rule ($\delta \equiv 0.01$, the first version) is $30\times$ the clean SER at a $3 \cdot 10^{-4}$ operating point; the relative rule with $\rho = 1$ and $\delta_{\min} = 2 \cdot 10^{-3}$ (five standard deviations of the pass fraction) follows the operating point.

Proposition 2 (residual gate under model mismatch)**.** *Write $\boldsymbol{y}_n = \hat{\boldsymbol{H}} \boldsymbol{s}_n + \boldsymbol{\nu}_n + \boldsymbol{m}_n$ with $\boldsymbol{m}_n$ the model-mismatch term. For decisions with error pattern $\boldsymbol{e}_n$, $r = \mathrm{mean} \left\| \hat{\boldsymbol{H}} \boldsymbol{e}_n \right\|^2 + \mathrm{mean} \|\boldsymbol{\nu}_n + \boldsymbol{m}_n\|^2 + O(N_d^{-1/2})$, so the ratio $r(\hat{\boldsymbol{s}}_{\mathrm{nn}})/r(\hat{\boldsymbol{s}}_{\mathrm{ref}})$ is monotone in the excess decision-error energy of the fast level and the mismatch only compresses it toward one.*

Table IV gives the measured constants. $L_\theta + L_a$ stays near one at every proximal weight: what the fast map forgets about its own past ($1 - L_\theta$) it inherits from the anchor (L_a), because the proximal step drives θ toward the anchor. The joint system is non-expansive rather than contractive, so the part of Theorem 1 claimed for the default receiver is the structural bound (9) with the measured Δ_f, checked in Section VI-G; the contraction branch is stated for completeness and is not certified by these probes. The same table shows that $\lambda = 1$ cuts the input sensitivity G_Φ of the fast level from 5.2 to 1.5: a stronger prior makes the per-block fit less steerable by the pilots.

V. EXPERIMENTAL SETUP

A fair test needs strong attackers, the receivers of the literature, several channel families and many seeds, with every receiver seeing the same realisation. Table V lists the scenarios.

TABLE V
EVALUATION SCENARIOS (3 SEEDS; E3 AND E2 AT 10 DB WITH 5).

Id	Channel	Attacker	Sweep
E2	SED cosine, Jakes 0.05, $\tanh$ PA, COST2100	none	SNR 4–16 dB
E3	SED cosine, 10 dB	A1 white-box and transfer; jamming	SIR -10 to 10 dB
E3snr	SED cosine	A1 white-box, SIR 0	SNR 4–16 dB
E3nl, E3cost	$\tanh$ PA; COST2100	A1 white-box	SIR -10 to 10 dB
E4	Jakes 0.05	A2 targeted, SIR 0	G3 on and off
E5	SED cosine, 400 blocks	A3, 5% energy, 300 blocks	$C_K \in \{8, 32, 128\}$, paired clean run
E7	SED cosine, 10 dB	gate-aware, weight 5, 50, 500	gate subsets
E8	SED cosine, 8×8	A1 white-box	SIR -10, 0, 10 dB
E9	Jakes 0.01, 0.05, 0.10	A1 white-box, SIR 0	Doppler
Ablation	SED cosine, 10 dB	A1 white-box, SIR 0	16 variants

Baselines. LMMSE on the least-squares pilot estimate (no adaptation; the model-based reference); online DeepSIC [5]; Meta-DeepSIC [6]; a drift-detection receiver that adapts only when a z-test on the pilot loss fires (the "adapt less" strategy); a MedBN-style receiver that replaces the adaptation gradient by a median of means over four pilot groups [9]; and a single-memory Titans receiver, the same delta-rule fast learner and surprise rate with $K = 1$ and no gates [14], which isolates the memory hierarchy from the fast rule. All DeepSIC-type receivers share the architecture, the pre-training and the 300 adaptation steps per block of Table II.

Protocol. Every receiver is run on the same channel realisation, noise and attack draw for a given seed (paired design); the attacker re-crafts its perturbation for each victim. Means over seeds are reported; for the main cells, 95% bootstrap confidence intervals over the (seed, block) pairs (10 000 resamples [23]) and two-sided Wilcoxon signed-rank tests [24] on the paired per-block differences, Holm-corrected within each table [25], are added. The attack window, the recovery criterion and the budget definition were fixed before the sweeps; the one change made after seeing results, the relative CRC slack, is reported together with its first, failed calibration (Section VI-J).

VI. RESULTS

The receivers are first compared under the availability attack; the same runs are then viewed from the gate's side; the SNR is varied; significance is tested; the receiver is stressed with the targeted, boiling-frog and gate-aware attackers; the certificate is checked; the receiver is dissected; and its transfer to other channels, a larger system and faster fading is tested.

A. Main Comparison

Table VI and Figs. 4–5 contain the main result. Under the white-box attack, every baseline is destroyed at SIR ≤ 0 dB (SER 0.59–0.98) and still loses 4–$6\times$ at 10 dB; Meta-DeepSIC and the MedBN-style receiver do not recover within 30 blocks (post-attack SER 0.19–0.20 against 0.06–0.07 before), reproducing the observation of [7]. The transfer attack is weaker against the baselines (0.20–0.36) but far from harmless, and it wrecks LMMSE at high budgets. The jamming control places the baselines' vulnerability in perspective: against every one of them, poisoning 200 pilot bits costs as much as jamming 50 000 data bits with the same energy (0.77 against 0.86 at 0 dB) and, unlike jamming, leaves persistent damage.

Against Citadel-Rx the attack is worthless at every budget and under both knowledge models. The attack-window SER (0.025–0.037) is at or below the receiver's own clean level in those blocks (Fig. 4a: the window covers the easy part of the fade, so the relevant comparison is with the dashed no-attack curve, from which the attacked run is indistinguishable), recovery is immediate, and the slow memory follows its no-attack trajectory (Fig. 4c). The attacker is reduced to jamming, which costs continuous power and leaves no trace. Before the attack, Citadel-Rx is 44% below online DeepSIC (0.040 against 0.071); Section VI-E examines that gain.

B. The Gate's Side

Table VII and Fig. 6 view the same runs from the gate. Rejection falls with the attacker's budget ($0.98 \rightarrow 0.86$) because a weak perturbation does less damage and more often stays inside the CRC slack; the blocks it leaks cost little, which is why the SER column of Table VI rises only from 0.026 to 0.034. False rejections are at most 1.7%. The CRC gate is the strongest single detector of the availability attack (AUC 0.999), the residual gate is close, and the prior gate is weak against A1, as designed. Under data jamming the residual ratio is inverted (the fast level's decisions fit the model better than the reference's; AUC 0.117 for G1 alone) and the CRC gate carries the verdict alone: the gates are not interchangeable, and $\gamma = \min$ inherits the best of them.

C. Degradation Across SNR

Fig. 7 reports the degradation metric of [7]. For every baseline the attacked SER stays at 0.45–0.91 over the whole 4–16 dB grid, so the extra SNR needed to recover the clean SER exceeds the grid everywhere. For Citadel-Rx, $\Delta_{\mathrm{dB}} \leq 0$ from 4 to 10 dB and ≈ 0 at 12 dB with the relative slack (0.0062 attacked against 0.0059 clean); at 14 and 16 dB the attacked window sits at 0.0018 and 0.0009 against clean 0.0008 and 0.0001. This is the limit of the defence: it holds the receiver near 10^{-3} under attack but does not preserve a 10^{-4} operating point, because the 5–10% of poisoned blocks that pass the gate set the floor. Closing it requires a rejection rate of one, not a different slack.

D. Significance and Seed Spread

Differences of a few SER points call for a significance test. Table VIII compares like with like: every pair of receivers sees the same channel, noise and attack draw, and the per-block differences are tested with Holm correction over the comparisons of each cell. Every difference in Tables VI and X is significant at $p < 0.001$, and the intervals exclude zero by a wide margin. Table IX adds the spread over five training seeds: the seed standard deviation of the attacked SER of Citadel-Rx is 0.002–0.003, two orders of magnitude below the effect. The Jakes clean cell spreads for every receiver because the realised fade differs by seed; the ranking is the same in all five seeds.

TABLE VI

SER IN THE ATTACK WINDOW (BLOCKS 30–69) UNDER A1 AT SIR $-10/-5/0/5/10$ DB ON THE 4×4 SED CHANNEL AT 10 DB; MEAN OVER SEEDS (5 FOR ONLINE DEEPSIC, META-DEEPSIC AND CITADEL-RX, 3 OTHERWISE). PRE: BLOCKS 0–29. POST AND RECOVERY (REC., BLOCKS; 30: NOT WITHIN THE RUN) AT SIR 0 DB. LMMSE HAS NO ADAPTATION MAP; ITS ATTACKED ROW IS THE TRANSFER ONE. SCRIPTS: `SWEEP.PY E3`, `ANALYZE.PY`.

Receiver	Pre	White-box A1	Post	Rec.	Transfer A1	Rec.	Jamming 0 dB
LMMSE	0.056	–	–	–	0.873 / 0.740 / 0.447 / 0.149 / 0.064	0	0.892
Online DeepSIC	0.071	0.973 / 0.938 / 0.772 / 0.475 / 0.277	0.088	11	0.352 / 0.357 / 0.347 / 0.295 / 0.210	10	0.855
Meta-DeepSIC	0.062	0.861 / 0.805 / 0.593 / 0.400 / 0.241	0.186	30	0.195 / 0.201 / 0.213 / 0.211 / 0.175	14	0.851
Drift-detection	0.077	0.975 / 0.942 / 0.776 / 0.459 / 0.265	0.097	10	0.345 / 0.359 / 0.337 / 0.277 / 0.199	7	0.858
MedBN-style	0.065	0.933 / 0.853 / 0.738 / 0.549 / 0.397	0.204	30	0.724 / 0.641 / 0.544 / 0.423 / 0.296	30	0.858
Single-memory Titans	0.068	0.975 / 0.939 / 0.788 / 0.486 / 0.276	0.075	6	0.363 / 0.368 / 0.354 / 0.290 / 0.200	6	0.861
Citadel-Rx (proposed)	**0.040**	**0.026 / 0.025 / 0.028 / 0.030 / 0.034**	**0.029**	**0**	**0.025 / 0.025 / 0.027 / 0.030 / 0.037**	**0**	0.848

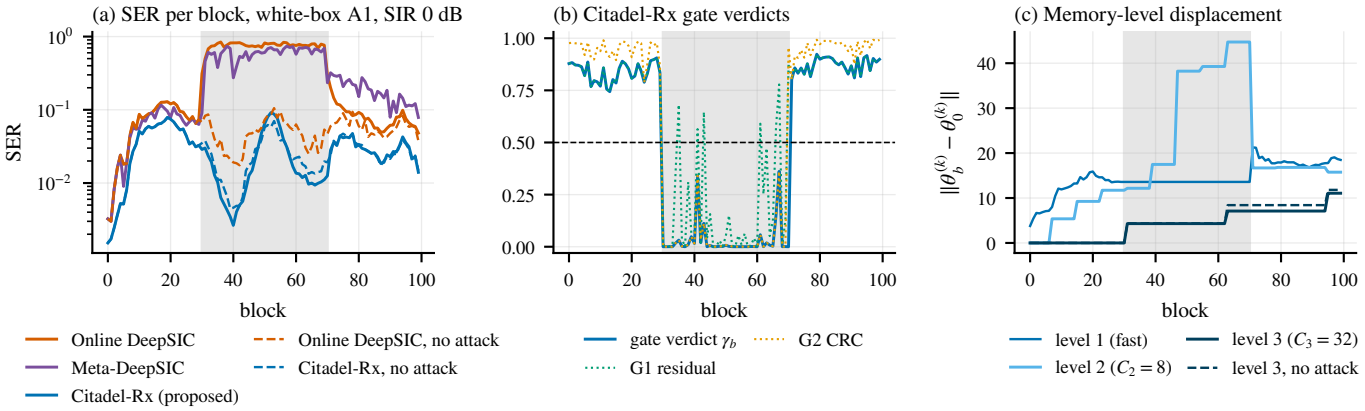


Fig. 4. One run (seed 0, white-box A1 at SIR 0 dB; shaded: attack window). (a) SER per block; dashed: the same receiver without attack. (b) Citadel-Rx gate verdicts: the CRC gate drops to zero on every poisoned block and the combined verdict rejects it. (c) Displacement of the three memory levels from their pre-trained value; level 2 absorbs the weights of the reverted fast states and returns, level 3 follows its no-attack trajectory. Script: `fig_paper.py`.

TABLE VII

GATE STATISTICS OF CITADEL-RX IN THE RUNS OF TABLE VI: REJECTION RATE OF POISONED BLOCKS / FALSE-REJECTION RATE OF CLEAN BLOCKS PER BUDGET, AND THE AREA UNDER THE ROC OF THE VERDICT AS A DETECTOR, POOLED OVER SEEDS AND BUDGETS.

Attacker	−10 dB	−5 dB	0 dB	5 dB	10 dB
White-box A1	0.98/0.013	0.98/0.007	0.97/0.003	0.95/0.000	0.86/0.000
Transfer A1	0.99/0.013	0.99/0.017	0.97/0.003	0.94/0.000	0.78/0.000

AUC under A1: γ 0.989; G1 alone 0.986, G2 alone 0.999, G3 alone 0.967
Jamming control: rejection 1.00, false rejection 0.000, AUC 1.000

E. Clean Performance

A defence is only useful if it costs nothing when there is no attack. Citadel-Rx costs less than nothing on three of the four channel families (Table X): it is the best receiver at every SNR from 4 to 12 dB on the cosine, Jakes and nonlinear channels, halving the SER of online DeepSIC at 10 dB. Pre-training does not explain the gain (pre-trained online DeepSIC: 0.074 against 0.069 in the pre-attack blocks), nor does the fast rule alone (single-memory Titans: 0.053); Section VI-H attributes it to the proximal anchor, i.e. to the slow memory acting as a prior for a fit from 100 pilot symbols. On COST2100, a nearly diagonal 4×4 channel on which LMMSE is close to optimal, Citadel-Rx matches LMMSE within 2–15% at 4–8 dB while every DeepSIC baseline is 1.5–2.3× worse than LMMSE. The clean gain is therefore channel-dependent, and nil where interference cancellation has nothing to cancel.

F. Targeted, Boiling-Frog and Gate-Aware Attackers

Table XI stresses the gates. The targeted attacker on a Jakes channel takes every user of online DeepSIC and Meta-DeepSIC down (window SER 0.44 and 0.35; the λ-weighted objective is not very selective), while Citadel-Rx stays at its clean 0.010 with 82% rejection, with or without the prior gate: G1 and G2 catch this attack on their own. The gate-aware attacker shows the value of several gates. Against the residual gate alone, a large penalty weight halves the rejection rate ($0.93 \to 0.30$) at the price of a weaker poison (SER $0.034 \to 0.074$): evading the residual test costs poisoning strength, as the design intended, but G1 alone is insufficient. Against G1 and G2 together, or all three, the best-effort attacker gains nothing (rejection 0.95–1.00, SER 0.024–0.027). The boiling-frog attacker passes the gates by construction (rejection 0) and buys one SER point over 300 blocks; its effect on the memory is the subject of the certificate.

G. The Certificate With Measured Constants

Fig. 8 and Table XII draw (9) against the paired adversarial/clean trajectories of the slowest level. The bound holds at every W for every period, is nearly tight at $C_K = 8$ (where the slow level is a running average of the fast level), and the measured slopes scale as $1 : 0.35 : 0.10$ against the $1 : 0.25 : 0.0625$ of pure $1/C_K$ ($\propto C_K^{-0.85}$ over a $16\times$ range); the deviation is Δ_f itself shrinking with C_K, because a slower anchor pulls the fast level back harder. The measured

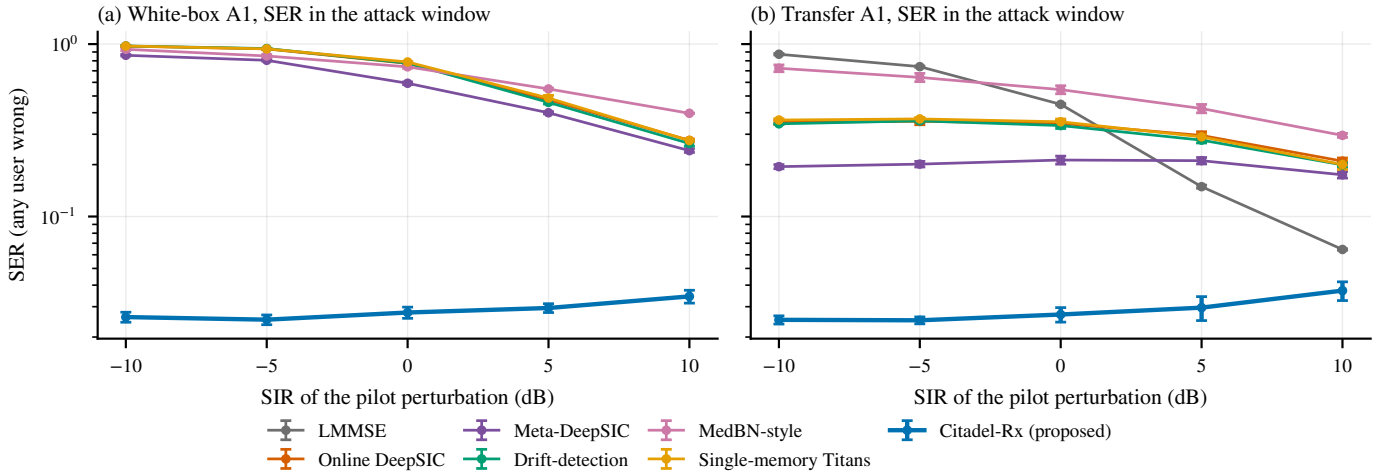


Fig. 5. SER in the attack window against the attacker's budget under (a) white-box and (b) transfer A1. Markers: seed means; bars: seed standard deviation. LMMSE appears in (b) only, since it has no adaptation map to attack through. Script: `fig_paper.py`.

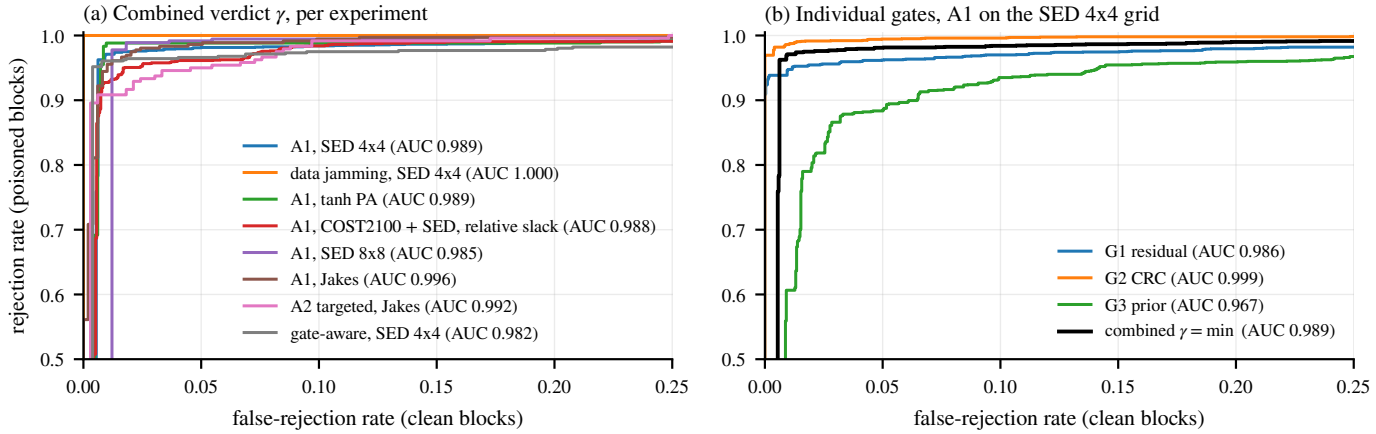


Fig. 6. Receiver operating characteristic of the gate verdict γ as a detector of poisoned blocks, pooled over seeds and budgets (axes zoomed to the upper-left corner): (a) combined verdict per experiment; (b) individual gates on the A1 grid. Script: `gate_roc.py`.

TABLE VIII
PAIRED COMPARISON ON (SEED, BLOCK) PAIRS: CITADEL-RX MINUS BASELINE, WITH 95% BOOTSTRAP INTERVAL AND HOLM-CORRECTED WILCOXON p (EVERY ENTRY $p < 0.001$). ATTACK CELLS USE THE WINDOW; CLEAN CELLS USE ALL 100 BLOCKS. FULL TABLES: `DOCS/STATS.MD`. SCRIPT: `STATS.PY`.

Citadel-Rx vs.	White-box A1, SIR 0 dB	Transfer A1, SIR 0 dB	Clean, cosine 10 dB	Clean, Jakes 10 dB
Online DeepSIC	−0.745 [−0.757, −0.731]	−0.320 [−0.331, −0.308]	−0.029 [−0.030, −0.028]	−0.0069 [−0.0075, −0.0063]
Meta-DeepSIC	−0.565 [−0.583, −0.546]	−0.186 [−0.198, −0.174]	−0.023 [−0.024, −0.021]	−0.0056 [−0.0061, −0.0051]
Drift-detection	−0.750 [−0.764, −0.734]	−0.310 [−0.325, −0.295]	−0.035 [−0.037, −0.033]	−0.0152 [−0.0166, −0.0139]
MedBN-style	−0.712 [−0.736, −0.686]	−0.517 [−0.539, −0.496]	−0.034 [−0.036, −0.032]	−0.0089 [−0.0097, −0.0080]
Single-memory Titans	−0.761 [−0.776, −0.745]	−0.328 [−0.342, −0.312]	−0.023 [−0.025, −0.021]	−0.0039 [−0.0046, −0.0032]
LMMSE	–	−0.420 [−0.432, −0.409]	−0.024 [−0.025, −0.022]	−0.0042 [−0.0049, −0.0035]

fast displacement (16–34) is of the size the geometric sum predicts from L_θ alone ($\epsilon G_\Phi/(1 - L_\theta) \approx 39$), so the anchor coupling does not compound even though $L_\theta + L_a > 1$.

H. Ablations

Each component is removed in turn (Table XIII). Three carry the result. The slow memory: with one level the gates fire but nothing uncorrupted is left to fall back on, and the receiver collapses like online DeepSIC (0.788); two levels halve the damage (0.050), three remove it. The fallback: reverting a rejected update protects the memory but leaves the data of the block to a poisoned fast level (0.245). The proximal anchor: it is the clean gain (0.039 against 0.058 without it), and the jump re-anchoring of [13] is no better than no anchoring; both also raise clean false rejections to 15–18% because the fast level drifts away from the memory it is compared with. The negative results are equally clear: the delta-rule decay and the surprise-driven step each buy 2–3% of clean SER; period alignment with the coherence structure does not matter on this channel; G3 is useless against A1, as predicted, its role being

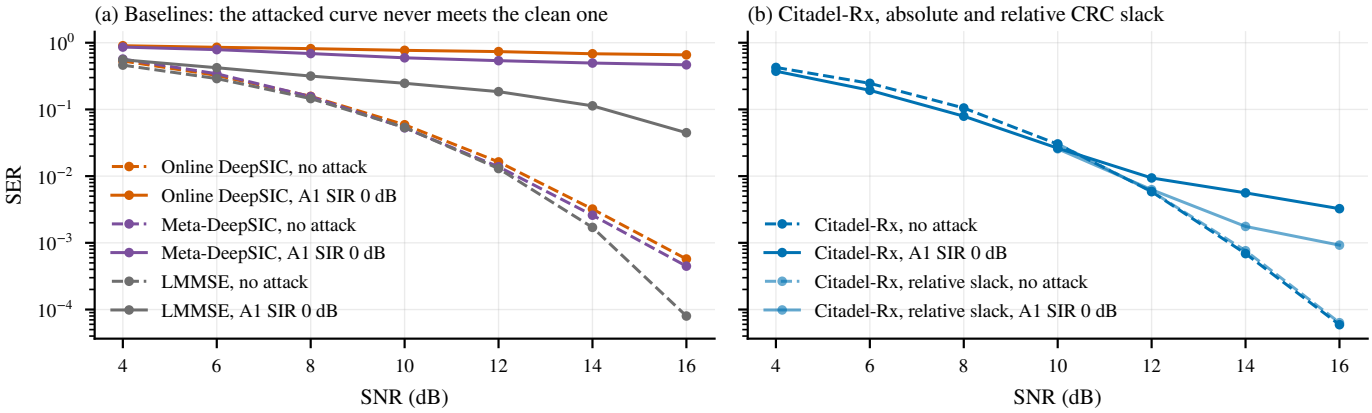


Fig. 7. SER against SNR without attack (dashed) and under white-box A1 at SIR 0 dB (solid), 3 seeds. (a) For every baseline the attacked curve never meets the clean one on the grid, so Δ_{dB} is unbounded. (b) Citadel-Rx with the absolute and the relative CRC slack; Δ_{dB} is the horizontal gap between the two curves. Script: `fig_paper.py`.

TABLE IX
HEADLINE CELLS WITH FIVE SEEDS: MEAN ± STANDARD DEVIATION OVER SEEDS.

Cell	Online DeepSIC	Meta-DeepSIC	Citadel-Rx
Clean, cosine 10 dB	0.0590 ± 0.0008	0.0527 ± 0.0008	$\mathbf{0.0302 \pm 0.0004}$
Clean, Jakes 10 dB	0.0128 ± 0.0058	0.0115 ± 0.0050	$\mathbf{0.0059 \pm 0.0028}$
A1 white-box, SIR −10	0.973 ± 0.003	0.861 ± 0.009	$\mathbf{0.026 \pm 0.002}$
A1 white-box, SIR 0	0.772 ± 0.013	0.593 ± 0.011	$\mathbf{0.028 \pm 0.002}$
A1 white-box, SIR 10	0.277 ± 0.005	0.241 ± 0.006	$\mathbf{0.034 \pm 0.003}$
A1 transfer, SIR 0	0.347 ± 0.013	0.213 ± 0.012	$\mathbf{0.027 \pm 0.003}$
Jamming, 0 dB	0.855 ± 0.001	0.851 ± 0.001	0.848 ± 0.001

TABLE X
SER WITHOUT ATTACK OVER 100 BLOCKS, 3 SEEDS (5 AT 10 DB ON THE COSINE AND JAKES CHANNELS). SCRIPTS: `SWEEP.PY E2`, `E2NL`, `E2COST`; `ANALYZE.PY`.

Receiver	Cosine 10 dB	Jakes 10 dB	tanh PA 10 dB	COST2100 8 dB
LMMSE	0.0537	0.0095	0.0562	**0.0041**
Online DeepSIC	0.0590	0.0128	0.0652	0.0096
Meta-DeepSIC	0.0527	0.0115	0.0595	0.0093
Drift-detection	0.0653	0.0206	0.0691	0.0140
MedBN-style	0.0644	0.0142	0.0605	0.0080
Single-memory Titans	0.0529	0.0092	0.0567	0.0102
Citadel-Rx (proposed)	**0.0302**	**0.0059**	**0.0331**	0.0047

A2; and whether a rejected update is reverted or kept makes no difference once the fallback is active.

I. Generalisation: Channels, System Size and Doppler

Table XIV repeats the main comparison on the nonlinear channel, the COST2100 traces, an 8×8 system and three Doppler rates. In every case, every baseline is destroyed at SIR ≤ 0 dB and Citadel-Rx stays at or below its pre-attack level. With eight users the clean gain grows (Citadel-Rx 0.102 against LMMSE 0.129 and online DeepSIC 0.204 over 100 clean blocks; interference cancellation matters more), Meta-DeepSIC and MedBN again never recover (post-attack 0.40–0.66), and false rejections rise to 3–5% because the eight-user fade has deeper per-block dips. Faster fading raises every receiver's clean SER and the gate's false alarms ($0.6 \rightarrow 3.3\%$), not the attacker's success: the attacked SER of Citadel-Rx is flat at 0.007 over a $10\times$ range of Doppler. Gate AUCs are

TABLE XI
STRESS ATTACKERS: SER IN THE ATTACK WINDOW (3 SEEDS; THE GATE-AWARE CELLS WITH WEIGHTS 50 AND 500 HAVE 1 SEED). REJ.: REJECTION RATE OF POISONED BLOCKS.

Attacker	Receiver	Pre	Window	Rej.
A2 targeted, Jakes 0.05, SIR 0 dB	Online DeepSIC	0.019	0.438	–
	Meta-DeepSIC	0.017	0.350†	–
	Citadel-Rx, G1+G2	0.010	0.010	0.82
	Citadel-Rx, G1+G2+G3	0.010	0.010	0.83
Gate-aware, SIR 10 dB	Citadel-Rx, no gates	0.039	0.531	0.00
	G1 only, weight 5	0.039	0.034	0.93
	G1 only, weight 50	0.039	0.044	0.85
	G1 only, weight 500	0.039	0.074	0.30
	G1+G2, weight 5	0.039	0.024	1.00
	G1+G2+G3, weight 5	0.039	0.026	0.99
	G1+G2+G3, weight 500	0.039	0.027	0.95
A3 boiling frog, 300 blocks	Citadel-Rx, $C_K = 8$	0.029	0.040‡	0.00
	Citadel-Rx, $C_K = 32$	0.031	0.039‡	0.00
	Citadel-Rx, $C_K = 128$	0.032	0.039‡	0.00

†post-attack 0.053, recovery 23 blocks. ‡post-attack 0.023.

TABLE XII
CERTIFICATE CHECK ON THE PAIRED A3 RUNS ($\eta^{(K)} = 0.5$, $p^{\mathrm{rej}} = 0$).

C_K	Δ_f (end of window)	$\max_W D_K(W)$/bound	Slope per block
8	33.4	0.94	0.084
32	17.2	0.53	0.029
128	15.9	0.26	0.0083

0.989 (nonlinear), 0.988 (COST2100, relative slack), 0.985 (8×8) and 0.996 (Jakes).

J. COST2100 and the Relative Slack

COST2100 at 10 dB is the one setting in which the first version of the receiver left its clean trajectory. With the absolute CRC slack ($\delta = 0.01$), the attack-window SER of Citadel-Rx was 0.0022/0.0032/0.0054/0.0122/0.0111 at SIR −10 to 10 dB against a clean 0.0003: 50–400× better than online DeepSIC, but with rejection falling to 0.64, because a one-point slack is 30× the clean SER and the attacker

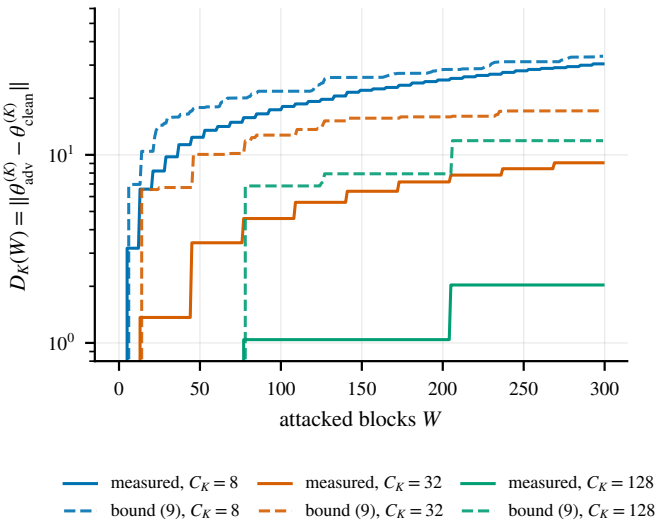


Fig. 8. Paired displacement of the slowest level under A3 (solid; 3 seeds) and the structural certificate (9) with the measured fast-level displacement (dashed) for $C_K = 8, 32, 128$ over 300 attacked blocks. Script: `certificate_check.py`.

TABLE XIII
ABLATIONS UNDER WHITE-BOX A1 AT SIR 0 DB, 3 SEEDS (COMPLETE RECEIVER ON THE SAME SEEDS). SCRIPT: `SWEEP.PY ABLATE`.

Configuration	Pre	Window	Rej.	False rej.
Complete receiver	0.039	**0.028**	0.97	0.003
$K = 1$ (gates, no slow level)	0.068	0.788	0.98	0.222
$K = 2$, $C = (1, 8)$	0.038	0.050	0.97	0.006
$K = 4$, $C = (1, 4, 16, 64)$	0.038	0.029	0.97	0.011
Periods misaligned, $C = (1, 3, 7)$	0.037	0.031	0.95	0.000
No gates	0.039	0.540	0.00	–
G1 only	0.039	0.031	0.97	0.000
G2 only	0.039	0.028	0.97	0.000
G3 only	0.039	0.507	0.08	0.000
Last layer adaptable	0.052	0.029	0.98	0.011
Plain GD fast rule	0.066	0.048	0.90	0.072
Adam without delta decay	0.040	0.030	0.97	0.006
Fixed step η	0.040	0.027	0.97	0.011
No anchor ($\lambda = 0$)	**0.058**	0.031	0.99	0.178
Jump re-anchor [13], $\beta = 0.25$	0.058	0.031	0.97	0.150
No fallback	0.040	**0.245**	0.97	0.011
Keep rejected update (no revert)	0.040	0.025	1.00	0.017

can pass G2 while doing damage. The relative slack of (10) brings the window to $0.0011/0.0012/0.0012/0.0015/0.0028$ (rejection $0.89/0.90/0.86/0.74/0.40$, no false rejections) and leaves the linear grid unchanged (0.025–0.029, rejection 0.93–1.00, false rejection $\leq 2\%$). A first calibration with a floor of 10^{-3} had fixed COST2100 but raised linear false rejections to 15–20%, because the floor sat inside the statistical resolution of the pass fraction; the floor of $2 \cdot 10^{-3}$ is five standard deviations of it. One further observation from this channel: LMMSE attacked through the surrogate is itself nearly immune here (0.0010 at SIR 0 dB), so on a nearly diagonal channel Citadel-Rx matches, rather than beats, the model-based receiver under attack (0.0013; paired difference $+0.0003$, interval $[-0.0005, +0.0010]$).

K. Cost

Per block on one CPU worker of the evaluation workstation (eight single-thread workers in parallel, 300 adaptation steps): LMMSE 0.07 s, online DeepSIC 4.0 s, drift-detection 9.2 s, Meta-DeepSIC 12.4 s, Citadel-Rx 12.2 s, single-memory Titans 14.4 s, MedBN-style 37.2 s. Citadel-Rx costs $3.1\times$ online DeepSIC and the same as Meta-DeepSIC; its overhead over the single-memory receiver is the gate (one LMMSE detection and one CRC check of the data block), while the levels cost memory (K copies of 15 628 parameters), not compute. The white-box attacker spends 3.2 s of PGD per poisoned block.

VII. DISCUSSION

Some conclusions are firmer than others; this section states what the evidence supports, what it does not, and what bounds the findings.

Against the attacks of [7] and their stronger variants, the robustness of the receiver is not in doubt: the attacked SER equals the clean one across budgets, knowledge models, channel families, system sizes and Doppler rates, with intervals and tests that leave no room for a seed artefact. The mechanism is visible in the records: the CRC gate drops to zero on poisoned blocks, the fast update is reverted, the data are detected with the slow memory, and the slow memory follows its no-attack trajectory. The ablation identifies which parts matter (the slow level, the fallback, and G1 or G2) and which do not (period alignment, the delta decay, the surprise rate, reverting versus keeping a rejected update). This is stated plainly because it is the finding most likely to be misread: the value of the receiver lies in its memory and its fallback, not in a cleverer fast learner.

The clean gain is a second contribution, but a conditional one. It is large where interference cancellation matters (44–50% lower SER than online DeepSIC on the SED channel, more with eight users) and nil on a nearly diagonal channel where LMMSE is already near optimal; the ablation attributes it to the proximal anchor, i.e. to a prior learned across blocks, which is why the jump re-anchoring of nested learning does not produce it.

The certificate is the structural bound (9) with a measured fast-level displacement; it holds at every window length and exhibits the $1/C_K$ mechanism. The contraction branch of Theorem 1 is not certified for the default receiver: the measured $L_\theta + L_a$ exceeds one at every proximal weight, and this was found to be structural rather than a probe artefact. A guarantee with no measured quantity requires a fast learner that is a contraction jointly in its state and its anchor, which the present design is not.

Two limits are stated as such. First, at clean SER near 10^{-4} the attacked window sits near 10^{-3}: the few poisoned blocks that pass the gate set a floor that a different slack does not move; a second-chance re-detection of accepted blocks would. Second, the CRC gate needs CRC-framed data; where the receiver sees only uncoded symbols, G1 and G3 remain and, by Table XIII, G1 alone defeats the non-adaptive attacker but not the gate-aware one.

The remaining assumptions bound the generality of the results. *Threat model:* the attacker perturbs the received pilots within an energy budget and knows the pilots; a jammer with a directional array or a spoofer of the data part lies outside the model. *Channel:* block fading with the synthetic, Jakes

TABLE XIV
TRANSFER TO OTHER CHANNELS, A LARGER SYSTEM AND FASTER FADING: SER IN THE ATTACK WINDOW UNDER WHITE-BOX A1, 3 SEEDS; PRE-ATTACK SER IN PARENTHESES; CITADEL-RX ON COST2100 WITH THE RELATIVE SLACK. SCRIPTS: `SWEEP.PY E3NL`, `E3REL`, `E8`, `E9`.

	tanh PA, 4×4		COST2100, 4×4		SED, 8×8		Jakes, $f_D T_b = 0.01/0.05/0.10$
Receiver	SIR -10	SIR 0	SIR -10	SIR 0	SIR -10	SIR 0	SIR 0
LMMSE (transfer)	0.841 (0.058)	0.421	0.063 (0.0001)	0.0010	0.983 (0.141)	0.793	0.067 / 0.067 / 0.103 (0.015–0.021)
Online DeepSIC	0.973 (0.079)	0.784	0.928 (0.0008)	0.636	0.998 (0.250)	0.971	0.719 / 0.719 / 0.717 (0.017–0.026)
Meta-DeepSIC	0.865 (0.067)	0.656	0.791 (0.0007)	0.345	0.972 (0.220)	0.934	0.496 / 0.501 / 0.531 (0.013–0.026)
Drift-detection	0.973 (0.083)	0.794	0.941 (0.0005)	0.630	0.997 (0.259)	0.971	0.728 / 0.738 / 0.717 (0.023–0.040)
MedBN-style	0.908 (0.069)	0.706	0.744 (0.0002)	0.505	0.995 (0.158)	0.923	0.711 / 0.692 / 0.691 (0.015–0.025)
Single-memory Titans	0.971 (0.075)	0.799	0.917 (0.0011)	0.627	0.998 (0.234)	0.969	0.738 / 0.749 / 0.736 (0.015–0.023)
Citadel-Rx (proposed)	**0.026** (0.042)	**0.026**	**0.0011** (0.0003)	**0.0013**	**0.080** (0.127)	**0.080**	**0.0066 / 0.0069 / 0.0075** (0.007–0.015)
rejection / false rej.	0.99 / 0.017	0.98 / 0.006	0.89 / 0.00	0.86 / 0.00	1.00 / 0.05	1.00 / 0.05	0.92 / 0.94 / 0.97; 0.006–0.033

and COST2100 families of the DeepSIC literature; no link-level simulation with a 3GPP channel model and LDPC coding was run. *Modulation:* QPSK with 100 pilot symbols per user. *Receiver:* DeepSIC; the levels and gates are agnostic to the per-user network, but no other backbone was tested. *Compute:* the reference detector and CRC check of the gate cost one LMMSE pass per block; the levels cost memory.

VIII. CONCLUSION

This paper presented a DeepSIC receiver that keeps its adaptable parameters at three time-scales, consolidates from fast to slow memory only through blocks that pass three label-free physics-consistency tests, pulls the fast level toward the slow projection, and detects a rejected block with the slow memory. Against the pilot-poisoning attacks that destroy online and meta-learned receivers, the attacked symbol error rate equals the clean one at every budget, on four channel families, two system sizes and three Doppler rates, with 86–99% of poisoned blocks rejected and at most 1.7% of clean ones; without attack the receiver halves the error rate of online DeepSIC where interference cancellation matters. The exposure of the slow memory to a bounded attack is divided by its period and multiplied by the miss rate of the gate, and the bound holds with measured constants. Two lessons follow. First, the defence lies in the memory, not in the learner: a single test-time memory with the same fast rule is as vulnerable as online DeepSIC. Second, a gate tied to the attacker's own objective, the CRC, is what a gate-aware attacker cannot evade without giving that objective up.

Several directions follow from the stated limits: a re-detection pass for accepted blocks to lift the 10^{-3} floor at high SNR; a fast learner that is a joint contraction, so that the second branch of the certificate becomes measurable; link-level simulation with LDPC coding and a 3GPP channel model, where the CRC gate becomes the transport-block CRC; and other backbones such as DeepRx-style convolutional receivers, to which the levels and gates apply unchanged.

DATA AND CODE AVAILABILITY

Code, configuration files, all 1 926 run records, the statistics and the scripts that regenerate every table, figure and number of this paper are available at https://github.com/abraheemrashid99-sys/Citradel-Rx. The COST2100 traces are those released with [6].

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Abraheem Rashid is with the Department of Computer Science, Brunel University London, London, U.K.